Caquan Palmer

Case Study 3 - Palmer-1.docx

Case Study 3

Computer Forensics I 25FA-CIS285B

Niagara Canvas CPF

Document Details

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7 Pages

491 Words

2,803 Characters

77% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Detection Groups

7 AI-generated only 77%
Likely AI-generated text from a large-language model.

0 AI-generated text that was AI-paraphrased 0%
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.



False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

Submission ID trn:oid:::1:3532417167

Precious.E01

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Case Study 3 This case study covered the full forensic workflow on the Precious.E01 image, from case creation through report generation. Indexed searches for hacking related term

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Case Study 3 Precious.E01

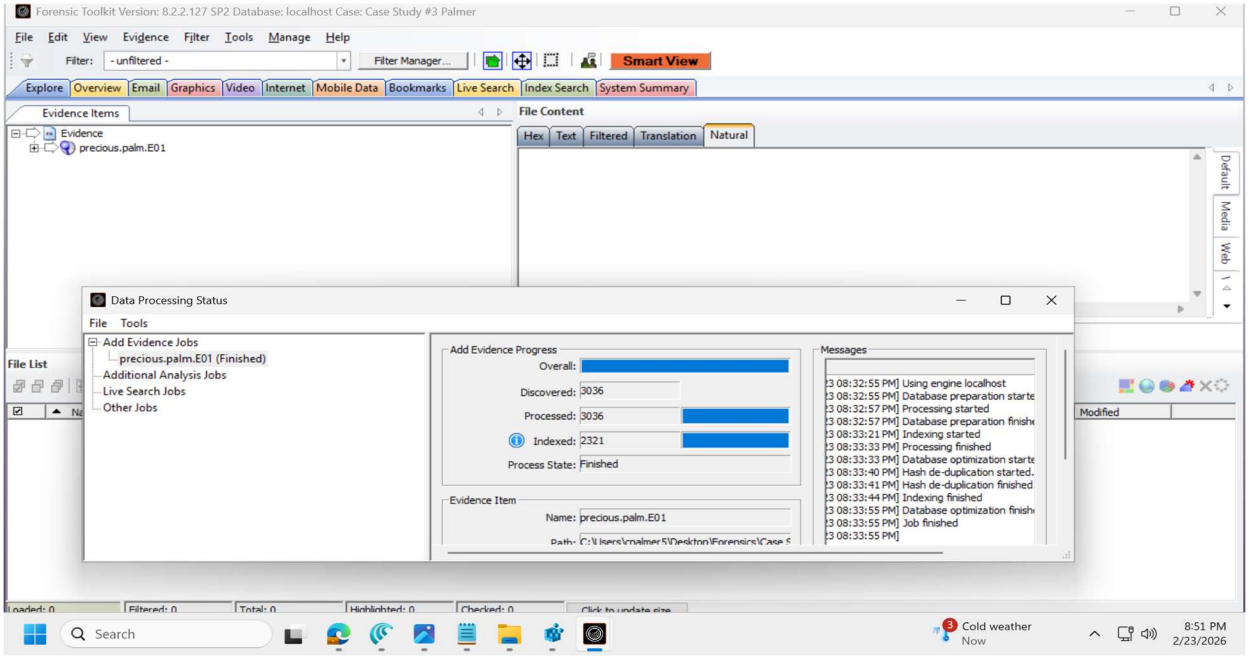
Introduction

This case study walks through a forensic examination of the Precious.E01 disk image using FTK. The work covers several core forensic tasks: loading and verifying evidence, identifying the file system, running indexed keyword searches, analyzing deleted files and email artifacts, attributing actions to specific user accounts, and performing data carving to recover content that the file system no longer tracks.

Each step was documented with bookmarks and screenshots to maintain an evidence chain, which is how a real forensic report would be structured. The point of the exercise is to show that deleted files, old emails, and user activity leave recoverable traces on a disk, and that tools like FTK can pull those traces together into something usable for an investigation.

1. Case Creation and Evidence Loading

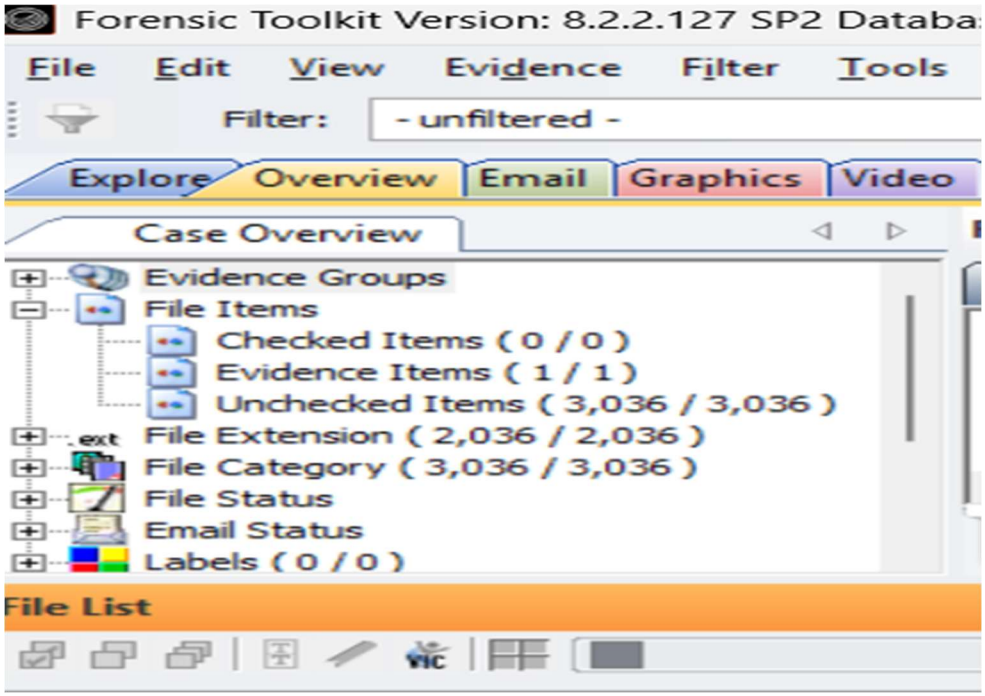
A new case was created in FTK and the image file `Precious.E01` was added as an evidence item. The image loaded successfully and was processed for analysis.



Evidence Tree showing Precious.E01 loaded

2. Total File Count

Total number of files: 3,030



3. File System

File system identified: NTFS

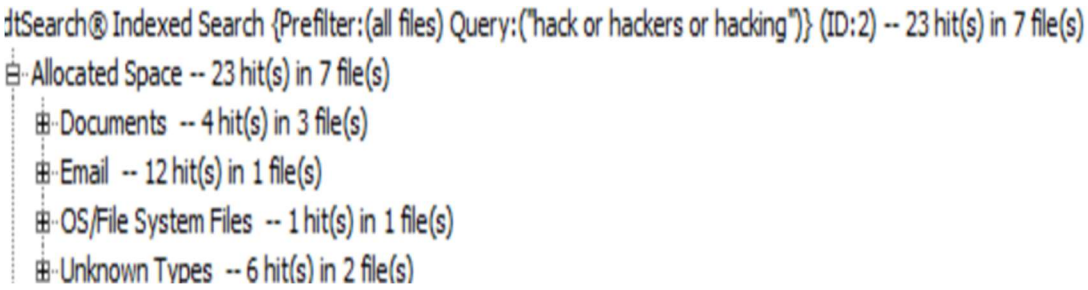
4. Indexed Search – Hacking Related Terms

An indexed search was conducted using the terms:

- hacking
- hackers
- hack

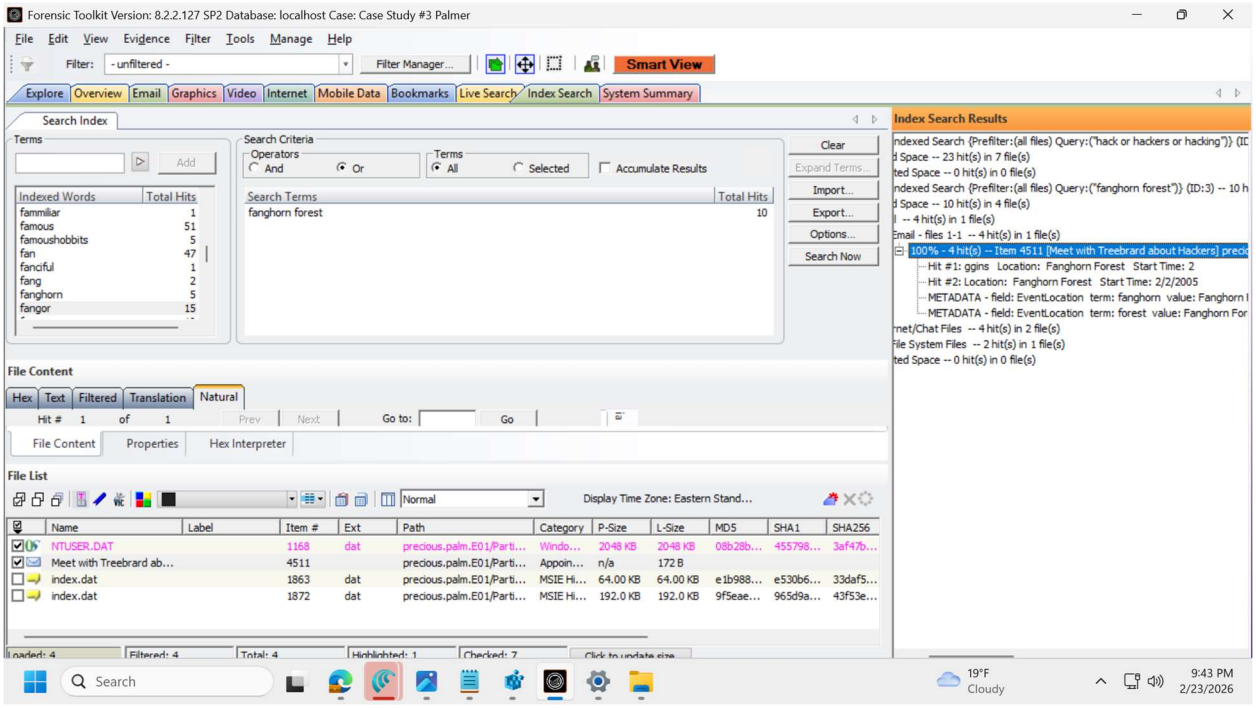
Relevant results were reviewed and bookmarked under a folder titled:

"Hacking Related Search"



5. Indexed Search – “Fanghorn forest”

An indexed search for “Fanghorn forest” identified an email appointment referencing the location.



The activity was attributed to the user account:

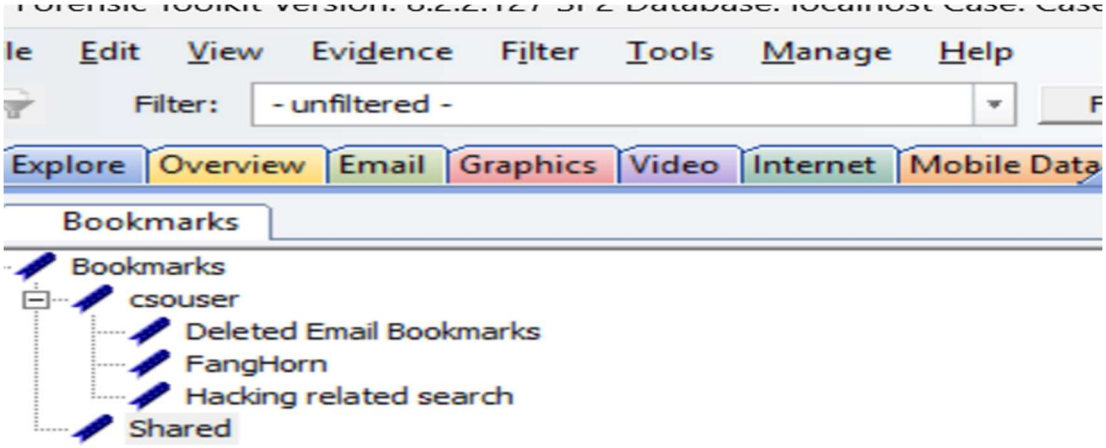
User_ ents and Settings/Frodo Baggins/Local Settings/Applicatio

associated user profile metadata.

6. Deleted Email Artifacts

Deleted email-related artifacts were identified using the Deleted file filter in FTK.

Relevant deleted email items were bookmarked under:

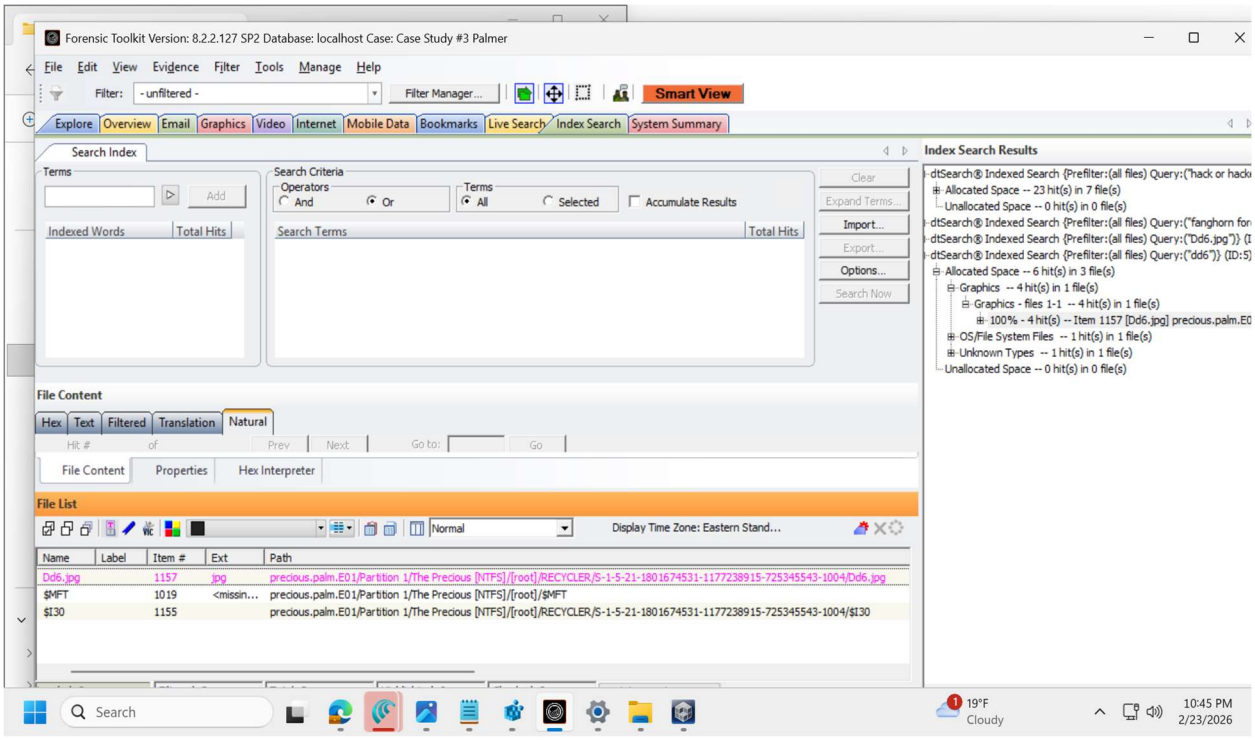


7. Deletion of Dd6.JPG

The file Dd6.JPG was identified as deleted.



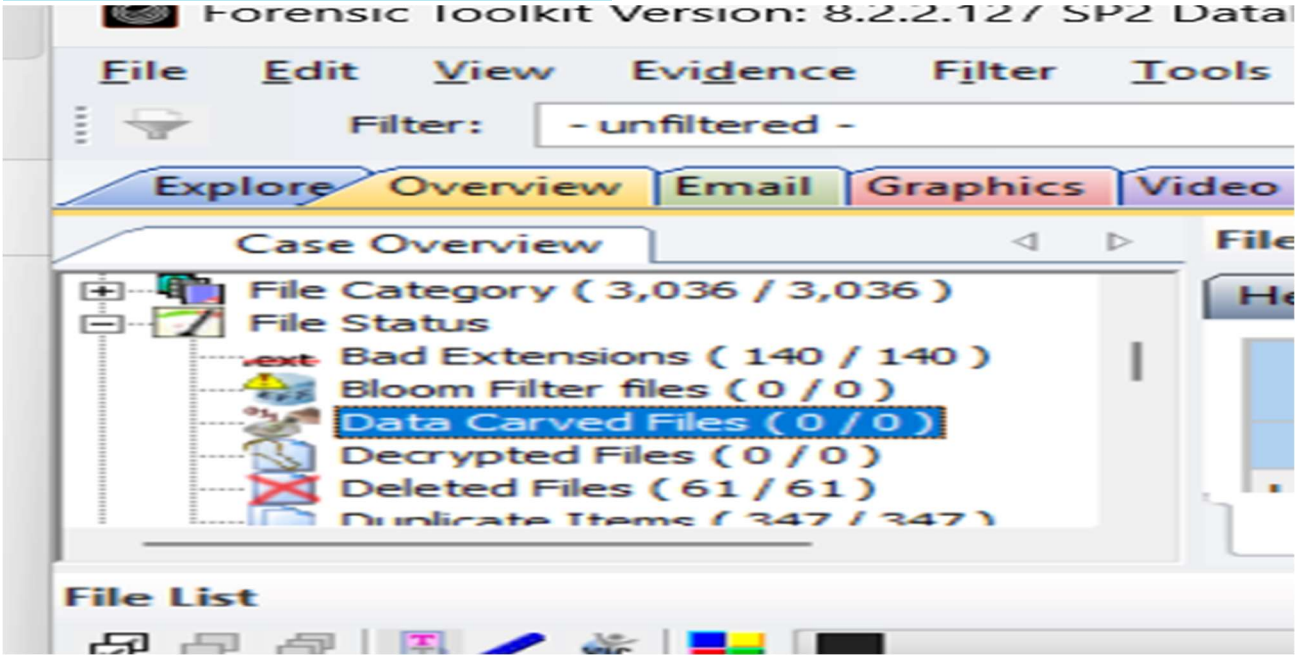
Deleted by user: artifact links the deletion to the associated user profile.



8. Automatic Data Carve

Automatic Data Carve was executed on the image.

File count before carving:3036
File count after carving: No carved Files Found



9. Report Generation

A report was generated in FTK including all bookmarks and bookmarked images. The report was exported in PDF format for submission.

Conclusion

The `Precious.E01` image was processed in FTK and examined. The forensic analysis of Precious.E01 involved indexing, keyword searches, deletion analysis, and data carving. After reviewing the SAM file in the registry viewer, it was determined that SID #(S-1-5-21-1645522239-343818398-839522115-1004 belonged to Frodo Baggins). Evidence artifacts were identified with bookmarks and screenshots used to document findings.

This case study covered the full forensic workflow on the Precious.E01 image, from case creation through report generation. Indexed searches for hacking related terms and "Fanghorn forest" returned results that tied specific activity to specific user accounts, showing how keyword searches can connect content to people.

Overall, the exercise reinforced that forensic tools can reconstruct user behavior well beyond what a casual user would expect to survive deletion, and that proper documentation through bookmarks and reports is what turns raw findings into presentable evidence